



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Project Background

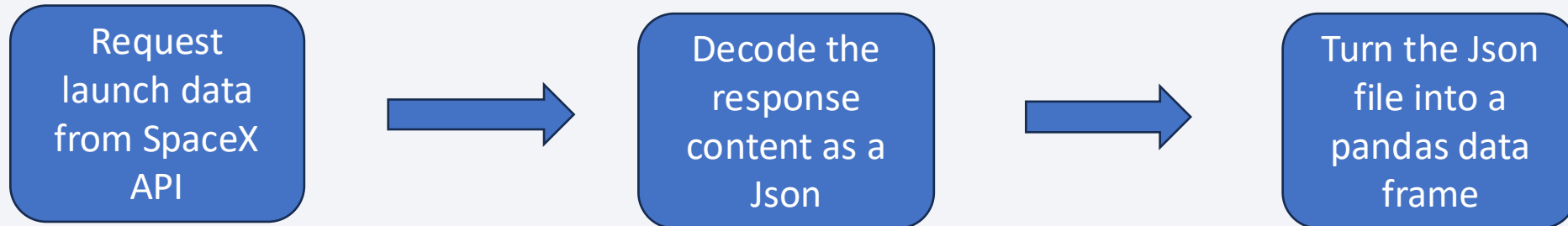
- The advertised cost for Falcon9 rocket launch is 62 million dollars based on SpaceX website. Other providers estimate a cost of 165 million dollars for each launch.
- The price difference is attributed to the reuse of the first stage by SpaceX which significantly reduces the cost.
- By determining if the first stage will land successfully, the cost for a launch can be estimated. This information can then be used by an alternate company to bid against SpaceX for a launch.

Data Collection

- Data are collected from SpaceX REST API and Web Scrapping Wikipedia

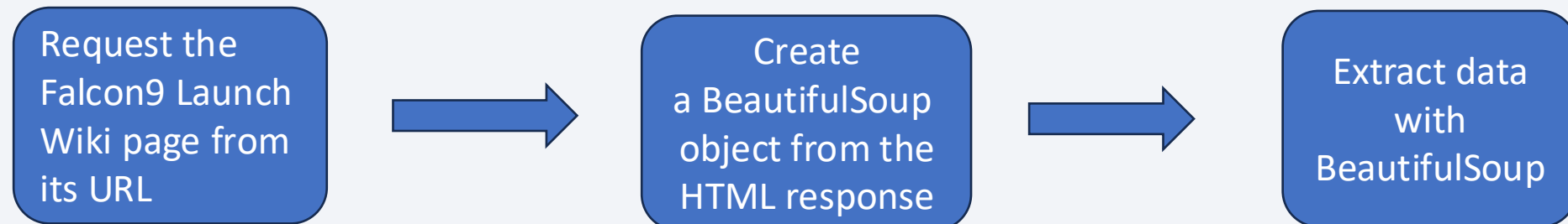
- SpaceX rocket launch data are obtained from SpaceX API using the following URL:

<https://api.spacexdata.com/v4/launches/past>



- Information such as launches, landing and payload weight are obtained by Web Scrapping Wikipedia using the URL:

<https://en.wikipedia.org/w/index.php?title=List of Falcon 9 and Falcon Heavy launches&oldid=1027686922>



Data Wrangling

1. Read the SpaceX dataset onto a data frame



2. Identify missing values in each attribute and identify numerical and categorical columns

```
df.isnull().sum()
```

```
df.dtypes
```



3. Calculate number of launches on each site

```
df['LaunchSite'].value_counts()
```



4. Calculate number and occurrence of each orbit

```
df['Orbit'].value_counts()
```

5. Calculate the number and occurrence of mission outcome of the orbits

```
landing_outcomes = df['Outcome'].value_counts()
```

landing_outcomes

Outcome	
True ASDS	41
None None	19
True RTLS	14
False ASDS	6
True Ocean	5
False Ocean	2
None ASDS	2
False RTLS	1
Name: count, dtype: int64	



6. Create a landing outcome label from Outcome column.

```
# landing_class = 0 if bad_outcome
# landing_class = 1 otherwise
landing_class = [0 if outcome in bad_outcomes else 1 for outcome in df['Outcome']]
```



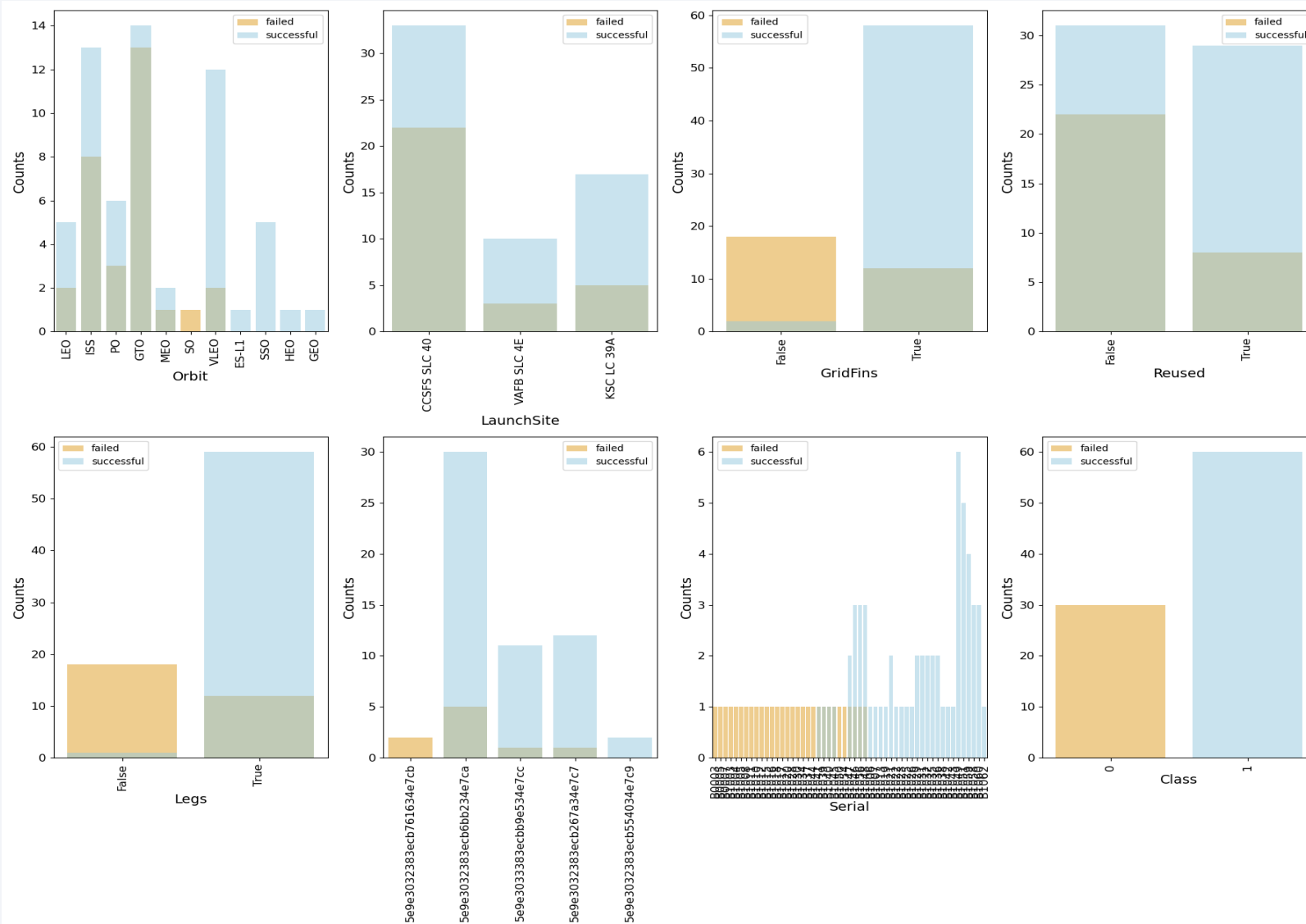
7. Create another column called class which shows the landing outcome. 0 or 1 based on a successful landing or not.

```
df['Class'] = landing_class
```

- GitHub URL of the completed Data Wrangling notebook:

[https://github.com/samalkapiy/IBM Applied Data Science capstone/blob/master/labs-jupyter-spacex-Data%20wrangling-v2.ipynb](https://github.com/samalkapiy/IBM_Applied_Data_Science_capstone/blob/master/labs-jupyter-spacex-Data%20wrangling-v2.ipynb)

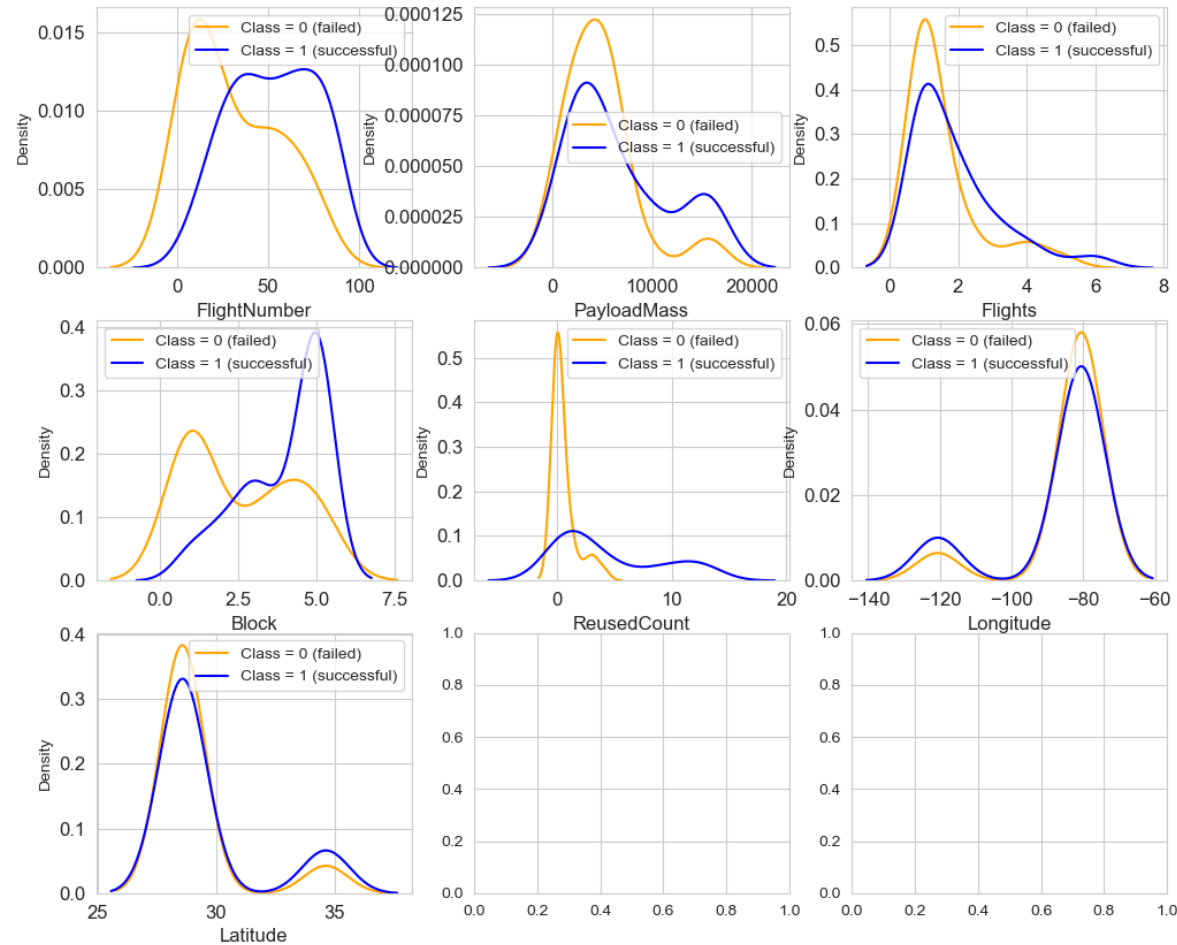
Exploratory Data Analysis (EDA)



- Bar plot for the categorical features.
- GridFins, Reused and Legs looks like features that can be used to differentiate failed and successful launches as they show different trends.

Exploratory Data Analysis (EDA)

Feature Density Plots

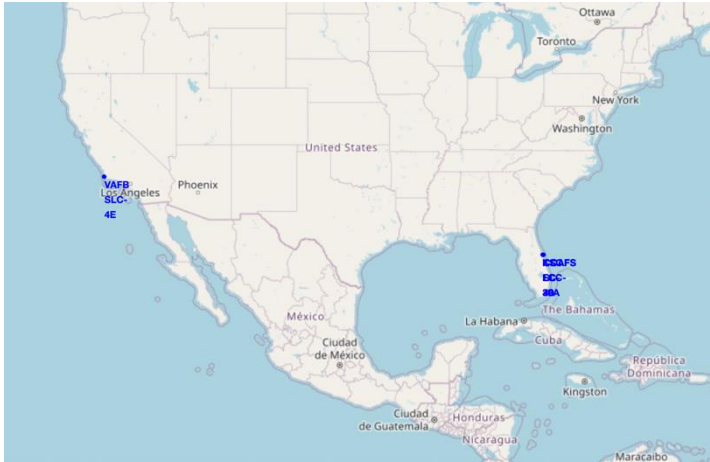


- Payload mass, Flights, Block, ReusedCount are features that show different trends for the two classes of failed and successful missions.

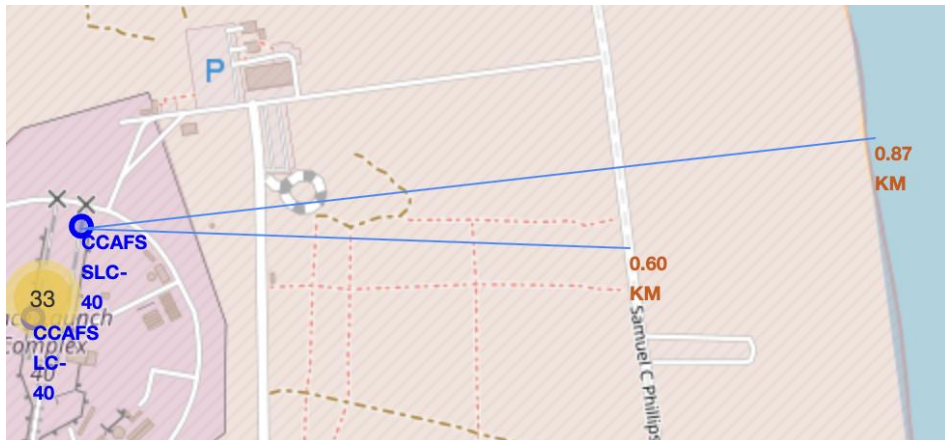


Launch Sites Proximities Analysis

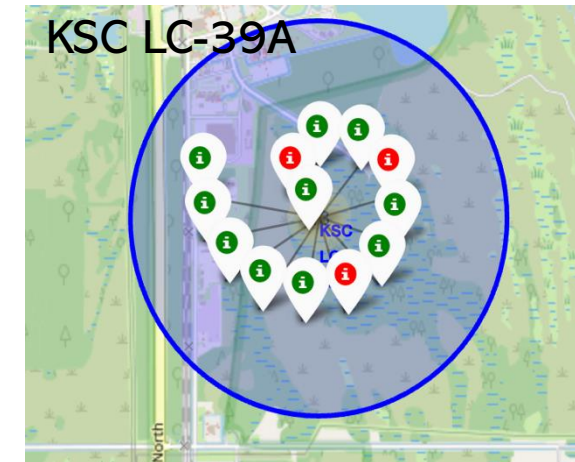
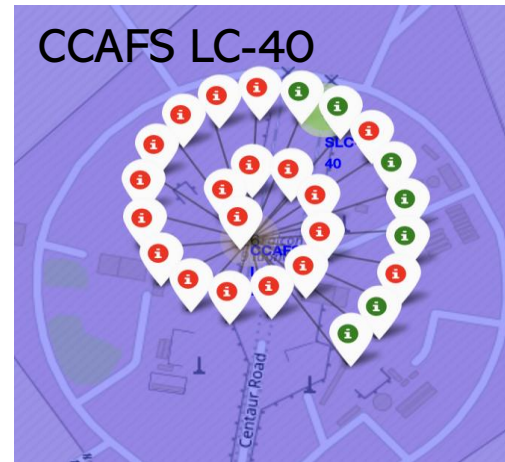
Folium Map SpaceX Launch Sites



Launch Site CCAFS SLC-40 and its Proximities



Successful and Failed Landings



The red markers denotes failed landings while green markers denote the successful landings.

- We can infer many information by indicating the distances from the Launch site to railroads, coastline, highways and nearby cities on the map.
- This information can be used in making decisions regarding transportation of rockets (rail roads used for transporting rocket parts) and risk assessment (having a city nearby would be a high risk).

GitHub URL of the completed Interactive map with Folium notebook:

https://github.com/samalkapiy/IBM_Applied_Data_Science_capstone/blob/master/lab-jupyter-launch-site-location-v2.ipynb



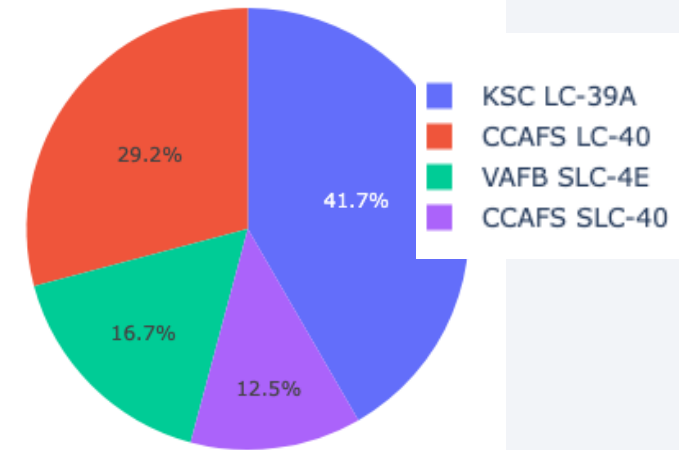
Section 4

Dashboard with Plotly Dash

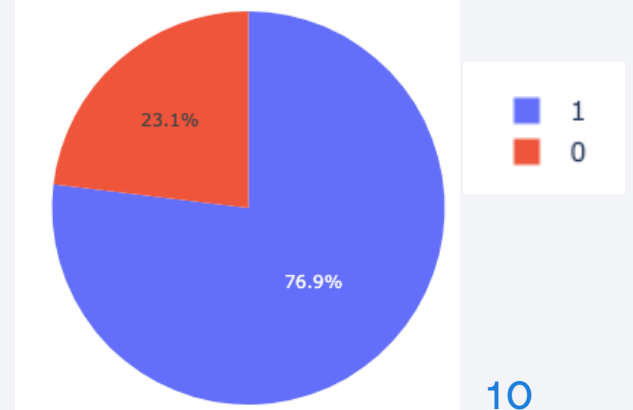
Dashboard with Plotly Dash

- Dashboard with a dropdown option where the user can select their desired Launch Site
- Pie charts are shown containing information such as the percentage of successful landings and failed landing based on the dropdown selection of the user.
- Rangeslider option allows the user to select a payload mass range and view information pertaining to the selected range
- Scatter plot shows Success vs Payload mass graph in the range selected by the user using Rangeslider option
- GitHub URL of the completed Plotly dash app:
https://github.com/samalkapiy/IBM_Applied_Data_Science_capstone/blob/master/spacex_dash_app.py

Successful launches by launch site



Successful launches for site KSC LC-39A



The site has 76.9% of successful landings and 23.1% of failed landings.



Section 4

Predictive Analysis (Classification)

Predictive Analysis (Classification)

1. Data preparation

Load Data, Standardize the data and split the data into training and testing sets.

2. Model Building

Select the Machine Learning Algorithm: XGBoost

Train the model on the training data set

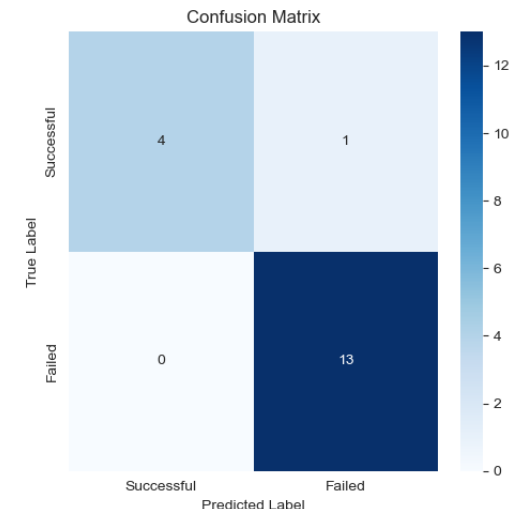
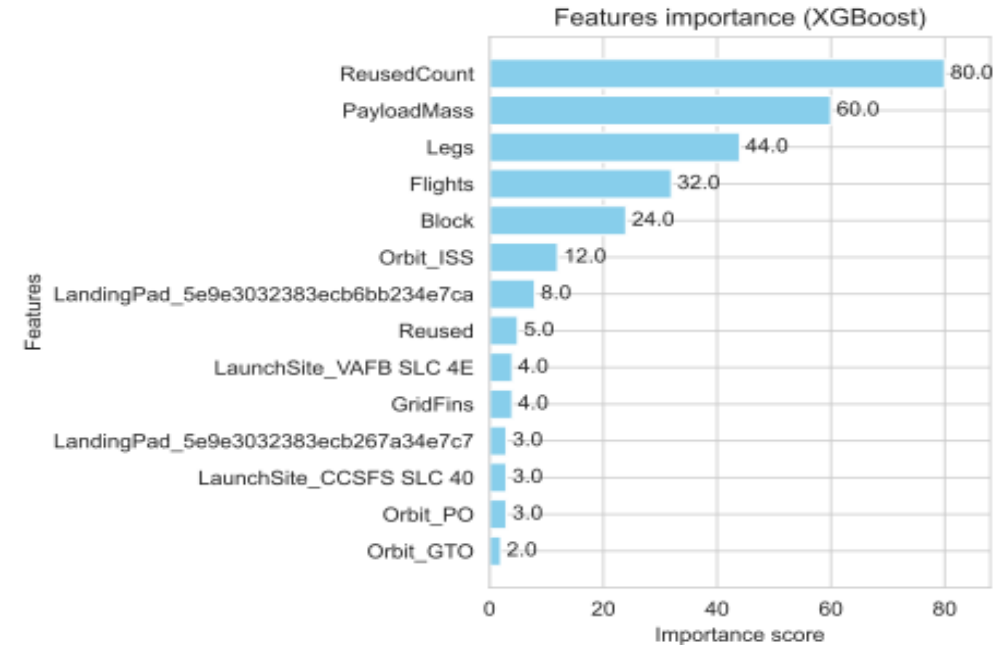
3. Model Evaluation

Evaluate baseline model

Plot the confusion matrix for the test dataset

Hyperparameter tuning

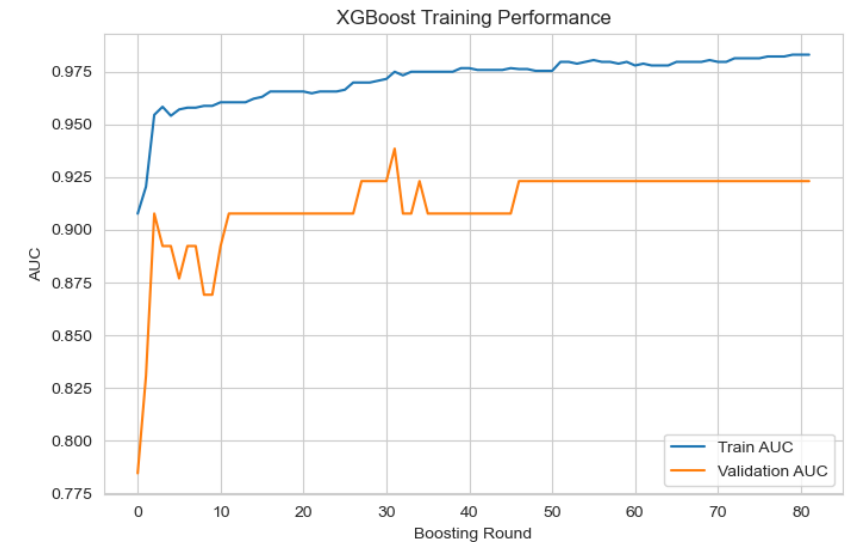
Calculate the accuracy of the model using the test dataset



Model Evaluation

- ML classifier used: XGBoost
- Accuracy metric for the baseline model: ROC-AUC((Area under the Receiver Operating Characteristic Curve)) : 0.92
- Accuracy of the model after hyperparameter tuning: ROC-AUC: 0.96
- Cross validation score: 0.89

Learning curve



Cost Estimation

- Using the hyper tuned model, cost estimation was conducted on the test dataset.
- Based on the model predictions, average cost of a launch after taking into account the successful landings of stage one:
 - 113.50 million dollars
- Total savings per launch : 51.50 million dollars
- Total savings for 18 launches (size of the test dataset): 927.00 million dollars

By using the model, the alternative company will be able to save 51.50 million dollars per launch.

Appendix

- All the notebooks, apps and data used for this project are added onto Github which can be accessed via this link:
https://github.com/samalkapiy/IBM_Applied_Data_Science_capstone/tree/master